MOBILE VALIDATION

1. Aim

To write and execute a Java program to validate a mobile number using exception handling.

2. Description

The program reads a mobile number from the user and checks whether it contains digits only and has exactly 10 digits. It uses NumberFormatException for non-digit input, InvalidLengthException for numbers shorter than 10 digits, and ArrayIndexOutOfBoundsException for numbers longer than 10 digits. Valid input displays a success message.

3. Software Requirements

• JDK 17 or later
• Java IDE or text editor (Eclipse, VS Code, or Notepad)
• Command Prompt / Terminal

4. Algorithm

1. Start.

2. Read the mobile number.

3. Check whether the number contains digits only.

4. If not, throw NumberFormatException.

5. If its length is less than 10, throw InvalidLengthException.

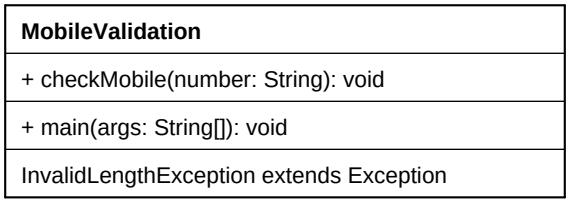
6. If its length is greater than 10, throw ArrayIndexOutOfBoundsException.

7. If the number has exactly 10 digits, display “Valid mobile number.”

8. Catch the exception and display the error message.

9. Stop.

5. UML Diagram



6. Source Code

```
package recordprograms;

import java.util.Scanner;

class InvalidLengthException extends Exception
{
    InvalidLengthException(String msg)
    {
        super(msg);
    }
}

public class MobileValidation
{
    static void checkMobile(String number)
            throws InvalidLengthException
    {
        if (!number.matches("\\d+"))
            throw new NumberFormatException("Digits only.");

        if (number.length() < 10)
            throw new InvalidLengthException("Number is too short.");

        if (number.length() > 10)
            throw new ArrayIndexOutOfBoundsException("Number is too long.");

        System.out.println("Valid mobile number.");
    }

    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter mobile number: ");
```

```
        String number = sc.nextLine();

        try
        {
            checkMobile(number);
        }
        catch (InvalidLengthException |
               NumberFormatException |
               ArrayIndexOutOfBoundsException e)
        {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

7. Output

Valid input

```
Enter mobile number: 9876543210
Valid mobile number.
```

Number too short

```
Enter mobile number: 12345
Error: Number is too short.
```

Number too long

```
Enter mobile number: 12345678901
Error: Number is too long.
```

Non-digit input

```
Enter mobile number: 98AB543210
Error: Digits only.
```

8. Result

Thus, the Java program to validate a mobile number using exception handling was written and executed successfully. It displays the appropriate message for valid and invalid mobile number inputs.